

CUSTOMER SHOPPING BEHAVIOR ANALYSIS REPORT

End-to-End Data Analytics Project

Tools: Python (JupyterLab) • PostgreSQL (pgAdmin) • Power BI

Dataset: 3,900 customer records • 18 features

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1. Executive Summary

This report documents a complete end-to-end data analytics project performed on a retail company's customer shopping behavior dataset. The project was executed in three stages: data preparation and cleaning in Python (JupyterLab), structured business query analysis in SQL (PostgreSQL via pgAdmin), and interactive visualization in Power BI. The goal was to answer the following overarching business question:

"How can the company leverage consumer shopping data to identify trends, improve customer engagement, and optimize marketing and product strategies?"

The analysis uncovered that male customers generate more than double the revenue of female customers, 80% of the customer base qualifies as Loyal, discounted customers still spend above average, and Clothing dominates revenue across all categories.

3,900 Total Records	\$59.76 Avg Purchase	3.75★ Avg Rating	80% Loyal Customers
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2. Problem Statement & Deliverables

A leading retail company observed shifting purchasing patterns across demographics, product categories, and sales channels. Management wanted to uncover which factors — discounts, reviews, seasons, payment preferences — drive consumer decisions and repeat purchases.

Business Questions Addressed

- **Q1:** Revenue comparison by gender
- **Q2:** High-spending customers who also used discounts
- **Q3:** Top 5 products by average review rating
- **Q4:** Average spend by shipping type (Standard vs. Express)
- **Q5:** Subscriber vs. non-subscriber spending comparison
- **Q6:** Products with highest discount application rates
- **Q7:** Customer segmentation — New, Returning, Loyal
- **Q8:** Top 3 most purchased products per category
- **Q9:** Repeat buyer subscription likelihood
- **Q10:** Revenue contribution by age group

Project Deliverables

Phase	Tool	Output
Data Preparation & EDA	Python / JupyterLab	Cleaned dataset, engineered features, loaded to PostgreSQL
SQL Analysis	PostgreSQL / pgAdmin	10 business queries with results

Phase	Tool	Output
Visualization	Power BI	Interactive dashboard with slicers
Reporting	Word / PDF	This report — findings & recommendations
Repository	GitHub	Python notebook, SQL file, .pbix dashboard

3. Dataset Overview

The dataset contains 3,900 purchase records across 18 columns covering customer demographics, transaction details, product categories, shipping behavior, discount usage, and review ratings.

Column	Type	Description
customer_id	INT	Unique identifier per customer
age	INT	Customer age (18–70)
gender	STR	Male or Female
item_purchased	STR	Specific product name
category	STR	Clothing, Accessories, Footwear, Outerwear
purchase_amount	INT	Transaction value in USD (\$20–\$100)
location	STR	US State
size	STR	S / M / L / XL
color	STR	Product color
season	STR	Spring, Summer, Fall, Winter
review_rating	FLOAT	Customer rating (2.5–5.0); had 37 nulls
subscription_status	STR	Yes / No
shipping_type	STR	Standard, Express, Free, Next Day Air, etc.
discount_applied	STR	Yes / No (identical to promo_code_used — dropped later)
previous_purchases	INT	Count of past purchases
payment_method	STR	PayPal, Credit Card, Cash, Venmo, etc.
frequency_of_purchases	STR	Weekly, Monthly, Annually, etc.

Raw Data — First 5 Rows (df.head())

[6]: df.head()

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied	Promo Code Used	Previous Purchases	Payment Method	Frequency of Purchases
0	1	55	Male	Blouse	Clothing	53	Kentucky	L	Gray	Winter	3.1	Yes	Express	Yes	Yes	14	Venmo	Fortnightly
1	2	19	Male	Sweater	Clothing	64	Maine	L	Maroon	Winter	3.1	Yes	Express	Yes	Yes	2	Cash	Fortnightly
2	3	50	Male	Jeans	Clothing	73	Massachusetts	S	Maroon	Spring	3.1	Yes	Free Shipping	Yes	Yes	23	Credit Card	Weekly
3	4	21	Male	Sandals	Footwear	90	Rhode Island	M	Maroon	Spring	3.5	Yes	Next Day Air	Yes	Yes	49	PayPal	Weekly
4	5	45	Male	Blouse	Clothing	49	Oregon	M	Turquoise	Spring	2.7	Yes	Free Shipping	Yes	Yes	31	PayPal	Annually

4. Data Preparation & EDA (Python / JupyterLab)

All data cleaning, transformation, and feature engineering was done in JupyterLab using the Pandas library. The cleaned data was then exported to a PostgreSQL database for SQL analysis.

4.1 Descriptive Statistics (df.describe())

The numeric columns were summarized using df.describe() to understand the range, spread, and distribution of key variables before cleaning.

```
[8]: df.describe()
```

	Customer ID	Age	Purchase Amount (USD)	Review Rating	Previous Purchases
count	3900.000000	3900.000000	3900.000000	3863.000000	3900.000000
mean	1950.500000	44.068462	59.764359	3.750065	25.351538
std	1125.977353	15.207589	23.685392	0.716983	14.447125
min	1.000000	18.000000	20.000000	2.500000	1.000000
25%	975.750000	31.000000	39.000000	3.100000	13.000000
50%	1950.500000	44.000000	60.000000	3.800000	25.000000
75%	2925.250000	57.000000	81.000000	4.400000	38.000000
max	3900.000000	70.000000	100.000000	5.000000	50.000000

- Average customer age: 44 years (range: 18–70)
- Average purchase amount: \$59.76 (range: \$20–\$100)
- Average review rating: 3.75 (range: 2.5–5.0)
- Average previous purchases: 25 (range: 1–50)

4.2 Missing Value Detection & Imputation

A null check using df.isnull().sum() revealed 37 missing values in the Review Rating column only.

```
[10]: df.isnull().sum()
```

```
[10]: Customer ID      0
      Age            0
      Gender         0
      Item Purchased  0
      Category       0
      Purchase Amount (USD)  0
      Location       0
      Size           0
      Color          0
      Season         0
      Review Rating   37
      Subscription Status  0
      Shipping Type   0
      Discount Applied  0
      Promo Code Used  0
      Previous Purchases  0
      Payment Method  0
      Frequency of Purchases  0
      dtype: int64
```

The missing values were filled using the category-level median rating, a more accurate approach than a global average, since review expectations differ across product types.

```
|: df["Review Rating"]=df.groupby("Category")["Review Rating"].transform(lambda x: x.fillna(x.median()))
```

After imputation, a second null check confirmed all 3,900 rows are complete with zero nulls.

```
[12]: df.isnull().sum()
[12]: Customer ID          0
      Age                0
      Gender              0
      Item Purchased      0
      Category            0
      Purchase Amount (USD) 0
      Location            0
      Size                0
      Color               0
      Season              0
      Review Rating       0
      Subscription Status 0
      Shipping Type       0
      Discount Applied    0
      Promo Code Used     0
      Previous Purchases  0
      Payment Method      0
      Frequency of Purchases 0
      dtype: int64
```

4.3 Column Standardization

Column names were converted to lowercase and spaces replaced with underscores for clean SQL and Python compatibility. The column `purchase_amount_(usd)` was also renamed to `purchase_amount`.

```
] : df.columns = df.columns.str.lower()
    df.columns = df.columns.str.replace(" ", "_")
    df = df.rename(columns = {"purchase_amount_(usd)": "purchase_amount"})
```

The updated `df.info()` confirms all 18 column names are clean, types are correct, and all 3,900 rows are non-null.

```
[14]: df.info()

<class 'pandas.DataFrame'>
RangeIndex: 3900 entries, 0 to 3899
Data columns (total 18 columns):
#   Column                Non-Null Count  Dtype
---  -
0   customer_id           3900 non-null   int64
1   age                   3900 non-null   int64
2   gender                3900 non-null   str
3   item_purchased        3900 non-null   str
4   category              3900 non-null   str
5   purchase_amount       3900 non-null   int64
6   location              3900 non-null   str
7   size                  3900 non-null   str
8   color                 3900 non-null   str
9   season                3900 non-null   str
10  review_rating         3900 non-null   float64
11  subscription_status   3900 non-null   str
12  shipping_type         3900 non-null   str
13  discount_applied      3900 non-null   str
14  promo_code_used       3900 non-null   str
15  previous_purchases    3900 non-null   int64
16  payment_method        3900 non-null   str
17  frequency_of_purchases 3900 non-null   str
dtypes: float64(1), int64(4), str(13)
memory usage: 548.6 KB
```

4.4 Feature Engineering

4.4.1 age_group Column

Customers were split into four equal-sized quartile groups using `pd.qcut()` to enable demographic analysis.

```
|: # create a columns age_groups
labels = ["young Adult", "Adult", "Middle-aged", "Senior"]
df["age_group"] = pd.qcut(df["age"], q=4, labels = labels)

|: df[["age", "age_group"]].head(10)
```

```
|:   age  age_group
0    55  Middle-aged
1    19  young Adult
2    50  Middle-aged
3    21  young Adult
4    45  Middle-aged
5    46  Middle-aged
6    63    Senior
7    27  young Adult
8    26  young Adult
9    57  Middle-aged
```

4.4.2 purchase_frequency_days Column

The text-based frequency labels (e.g., 'Weekly', 'Fortnightly') were mapped to numeric day values to enable quantitative frequency analysis.

```

: # Create column purchase_frequency_days
frequency_mapping = {
    "Fortnightly" : 14,
    "Weekly" : 7,
    "Monthly" : 30,
    "Quarterly" : 90,
    "Bi-Weekly" : 14,
    "Annually" : 365,
    "Every 3 Months" : 90
}
df["purchase_frequency_days"] = df["frequency_of_purchases"].map(frequency_mapping)

: df[["purchase_frequency_days", "frequency_of_purchases"]]

:

```

	purchase_frequency_days	frequency_of_purchases
0	14	Fortnightly
1	14	Fortnightly
2	7	Weekly
3	7	Weekly
4	365	Annually
...	...	...
3895	7	Weekly
3896	14	Bi-Weekly
3897	90	Quarterly
3898	7	Weekly
3899	90	Quarterly

3900 rows x 2 columns

4.5 Redundant Column Removal

A check confirmed that `discount_applied` and `promo_code_used` were 100% identical across all 3,900 rows. `promo_code_used` was dropped to eliminate redundancy, reducing the feature count to 17 columns.

```
df[["discount_applied", "promo_code_used"]]
```

	discount_applied	promo_code_used
0	Yes	Yes
1	Yes	Yes
2	Yes	Yes
3	Yes	Yes
4	Yes	Yes
...	...	...
3895	No	No
3896	No	No
3897	No	No
3898	No	No
3899	No	No

3900 rows x 2 columns

```
(df["discount_applied"] == df["promo_code_used"]).all()
```

```
np.True_
```

```
df = df.drop("promo_code_used", axis=1)
```


4.6 Loading to PostgreSQL via SQLAlchemy

After all cleaning and feature engineering steps, the finalized DataFrame was exported to a local PostgreSQL 18 database (customer_behavior) using SQLAlchemy and psycopg2. The data was written to a table named customer for SQL analysis.

```
from sqlalchemy import create_engine

# 1. Force the correct variables
username = "postgres"
password = "mymac"
host = "localhost"
port = "5432" # Explicitly targeting your PostgreSQL 18 instance
database = "customer_behavior"

# 2. Re-build the engine in the exact same cell execution
connection_string = f"postgresql+psycopg2://{username}:{password}@{host}:{port}/{database}"
engine = create_engine(connection_string)

# 3. Write data to the table
table_name = "customer"
df.to_sql(table_name, engine, if_exists="replace", index=False)

print(f"Data successfully loaded into table '{table_name}' in database '{database}'.")
```

Data successfully loaded into table 'customer' in database 'customer_behavior'.

5. Data Analysis (SQL / PostgreSQL — pgAdmin)

Ten SQL queries were written and executed in pgAdmin against the customer table in the customer_behavior PostgreSQL database. Queries cover aggregations, subqueries, CTEs, and window functions.

Query 1 — Revenue by Gender

Compared total revenue generated by male vs female customers.

	gender text 	revenue numeric 
1	Female	75191
2	Male	157890

Insight: Male customers generate more than double the revenue of female customers (\$157,890 vs \$75,191). This may indicate a larger male customer base or higher average basket size, both worth investigating through further segmentation.

Query 2 — High-Spending Discount Users

Identified customers who used discount but still spent above the average purchase amount.

	customer_id bigint	purchase_amount bigint
1	2	64
2	3	73
3	4	90
4	7	85
5	9	97
6	12	68
7	13	72
8	16	81
9	20	90
10	22	62
11	24	88
12	29	94
13	32	79
14	33	67
15	35	91
16	37	69
Total rows: 839		Query complete 00:00:00.104

Figure 17: Q2 Result — 839 customers used discounts and still spent above average (\$59.76)

Insight: 839 customers (~22%) used discounts yet spent above the overall average. This proves discounts attract high-intent buyers, not just budget shoppers, a strong signal that discount campaigns do not cannibalize revenue per transaction.

Query 3 — Top 5 Products by Review Rating

Found products with the highest average review rating.

	item_purchased text	average numeric
1	Gloves	3.86
2	Sandals	3.84
3	Boots	3.82
4	Hat	3.80
5	Skirt	3.78

Insight: Gloves, Sandals, and Boots lead in customer satisfaction ratings. These should be featured prominently in marketing campaigns and homepage placements to drive conversion on high-satisfaction products.

Query 4 — Average Spend by Shipping Type

Compare the average Purchase Amounts between Standard and Express Shipping.

	shipping_type text	average numeric
1	Standard	58.46
2	Express	60.48

Insight: Express shipping customers spend slightly more on average. This suggests premium shipping correlates with higher-value purchases. Offering free Express upgrade above a spend threshold (e.g., \$80+) could incentivize larger basket sizes.

Query 5 — Subscriber vs. Non-Subscriber Spending

Compare average spend and total revenue between subscribers and non-subscribers.

	subscription_status text	avg_spend numeric	total_revenue numeric	count bigint
1	Yes	59.49	62645	1053
2	No	59.87	170436	2847

Insight: Average spend is nearly identical between subscribers (\$59.49) and non-subscribers (\$59.87) — meaning subscription status does not change spending behavior. However, non-subscribers make up 73% of the base, representing a massive untapped subscription conversion opportunity.

Query 6 — Products with Highest Discount Rates

Query calculated the percentage of purchases per product where a discount was applied ($100 * \text{SUM}(\text{CASE WHEN discount_applied} = \text{'Yes'}) / \text{COUNT}(*))$, ordered descending, limited to top 5.

	item_purchased text	discount_rate numeric
1	Hat	50.00
2	Sneakers	49.00
3	Coat	49.00
4	Sweater	48.00
5	Pants	47.00

Insight: Certain products attract discount usage consistently. These items can be targeted with tiered loyalty perks or bundle deals instead of blanket discounts — protecting margin while maintaining engagement.

Query 7 — Customer Loyalty Segmentation (CTE)

Segment customers into New, Returning, and Loyal based on their total number of previous purchases, and show the count of each segment

	customer_segment text	number_of_customers bigint
1	Loyal	3116
2	Returning	701
3	New	83

Insight: An extraordinary 80% of customers qualify as Loyal (11+ purchases). Only 83 customers are truly New. This confirms strong retention, the growth strategy should shift toward maximizing loyalty value and subscription conversion rather than acquisition.

Query 8 — Top 3 Products per Category (Window Function)

Identified top 3 most purchased products within each category?

	item_rank bigint	category text	item_purchased text	total_orders bigint
1	1	Accessori...	Jewelry	171
2	2	Accessori...	Sunglasses	161
3	3	Accessori...	Belt	161
4	1	Clothing	Blouse	171
5	2	Clothing	Pants	171
6	3	Clothing	Shirt	169
7	1	Footwear	Sandals	160
8	2	Footwear	Shoes	150
9	3	Footwear	Sneakers	145
10	1	Outerwear	Jacket	163
11	2	Outerwear	Coat	161

Insight: Clothing is led by Blouse and Pants (171 orders each). Accessories by Jewelry (171). Footwear by Sandals (160). Outerwear by Jacket (163). These top items should be prioritized for inventory planning and featured product placement.

Query 9 — Repeat Buyers & Subscription Likelihood

Checked whether customers with >5 purchases are more likely to subscribe

	subscription_status text	repeat_buyers bigint
1	No	2518
2	Yes	958

Insight: Among customers with more than 5 purchases, 2,518 are not subscribed, 72% of repeat buyers. This is the single most actionable finding: a massive, high-purchase-frequency segment that has not yet converted to a subscription. A targeted campaign to this group could significantly grow subscription revenue.

Query 10 — Revenue by Age Group

Checked total revenue contribution of each age group

	age_group text 🔒	revenue numeric 🔒
1	young Adult	62143
2	Middle-aged	59197
3	Adult	55978
4	Senior	55763

Insight: Revenue is distributed nearly evenly across all four age groups (Young Adult, Adult, Middle-aged, Senior) with Young Adults slightly ahead. This indicates the product range has broad cross-generational appeal, marketing should not over-index on any single demographic.

6. Visualization & Insights (Power BI Dashboard)

An interactive Power BI dashboard was built to allow non-technical stakeholders to explore the data visually. The dashboard responds in real time to slicer selections for Subscription Status, Gender, Category, and Shipping Type.



Dashboard Components

- **KPI Cards (top):** Number of Customers (3.9K) | Average Purchase Amount (\$59.76) | Average Review Rating (3.75)
- **Donut Chart:** Subscription status split — 2.85K (73%) without subscription, 1.05K (27%) subscribed
- **Revenue by Category (bar):** Clothing leads at ~\$100K, followed by Accessories ~\$70K, Footwear, then Outerwear
- **Sales by Category (bar):** Clothing dominates with ~1,750 transactions; Accessories second at ~1,200
- **Revenue by Age Group (horizontal bar):** Young Adults generate slightly more, all groups within 10% of each other
- **Sales by Age Group (horizontal bar):** Consistent spread — broad cross-demographic demand confirmed
- **Slicers (right panel):** Subscription Status, Gender, Category (4 options), Shipping Type (6 options) — all interactive

7. Key Findings

Revenue & Demographics

- Male customers account for ~68% of total revenue (\$157,890 vs. \$75,191 for female).
- Revenue is nearly equal across all four age groups — no single age segment dominates.
- Clothing is the highest-revenue and highest-volume category by a clear margin.

Pricing & Discount Behavior

- 839 customers (22%) used discounts yet still spent above the average — discounts attract high-intent buyers.
- Subscribers and non-subscribers have nearly identical average spend (~\$59.50 vs. ~\$59.87).
- Express shipping customers spend \$2 more on average than Standard shipping customers.

Product Performance

- Gloves, Sandals, and Boots are the highest-rated products by customer review.
- Top sellers per category: Clothing → Blouse/Pants | Accessories → Jewelry | Footwear → Sandals | Outerwear → Jacket.

Customer Loyalty & Retention

- 80% of customers are Loyal (11+ previous purchases) — an exceptionally strong retention rate.
- Only 83 customers (2%) are New, confirming a mature, retention-focused base.
- 73% of customers (including thousands of repeat buyers) have not subscribed — major conversion opportunity.
- 2,518 repeat buyers (>5 purchases) are not subscribed — the highest-ROI target for subscription campaigns.

8. Business Recommendations & Next Steps

1. Subscription Conversion Campaign (Highest Priority)

2,518 repeat buyers with >5 purchases have never subscribed. This is the most actionable segment in the entire dataset.

- Deploy targeted email/SMS campaign to non-subscribed customers with 5+ purchases, offering first-month free or exclusive subscriber discount.
- Highlight subscriber-only benefits (free shipping, early access, loyalty points) in post-purchase flows.
- Set a measurable target: convert 20% of the 2,847 non-subscribers within 6 months.

2. Gender-Targeted Marketing

The female revenue gap (\$75K vs. \$157K) most likely reflects underrepresentation rather than lower individual spend.

- Audit current ad targeting — confirm whether female reach is proportionally invested.
- Design category-specific campaigns (Accessories, Footwear) aimed at female demographic segments.
- A/B test gender-personalized email subject lines and landing pages to measure conversion lift.

3. Product Spotlight Strategy

Top-rated products (Gloves, Sandals, Boots) and top-selling products (Blouse, Jewelry, Jacket) are not always the same.

- Feature top-rated products on the homepage hero banner and category landing pages to convert quality satisfaction into more sales.
- Investigate why high-volume items (Blouse, Pants) are not in the top-rated list — potential quality or expectation gap.
- Bundle Outerwear (lowest category revenue) with top-rated items to improve cross-category basket size.

4. Smart Discount Policy

Since discount users still spend above average, blanket discount removal would be counterproductive.

- Shift from blanket discounts to tiered incentives: larger discounts on higher basket sizes, smaller discounts on low-value orders.
- Consolidate `discount_applied` and `promo_code_used` into a single system (they were 100% identical — operational redundancy).
- Test loyalty-point rewards as an alternative to price discounts for the Loyal segment.

5. Shipping & Spend Optimization

- Introduce free Express shipping above a spend threshold (e.g., \$75+) to encourage higher basket sizes.
- Test a shipping-type recommendation engine that suggests Express for high-value items at checkout.

6. Loyalty Tier Program

- Introduce a formal 3-tier loyalty program (Silver/Gold/Platinum) to recognize and deepen Loyal customer engagement.
- Design a targeted nudge campaign for the 701 Returning customers (2–10 purchases) with a bonus incentive to reach Loyal status.
- Track customer lifetime value (CLV) quarterly as the primary retention KPI going forward.

9. Conclusion

This project demonstrated a complete data analytics pipeline, from raw CSV data through Python cleaning, PostgreSQL business querying, Power BI visualization, and strategic reporting. The retail company is operating from a position of strength with a highly loyal customer base and consistent spending patterns across all demographics and product categories.

The most critical findings point to three immediate opportunities: subscription conversion of the 2,518 non-subscribed repeat buyers, gender-targeted marketing to close the revenue gap, and product promotion aligned with customer satisfaction ratings rather than sales volume alone.

The combination of Python, SQL, and Power BI provides a scalable, repeatable analytics framework that can be refreshed as new transaction data flows in, enabling the business to make evidence-based decisions on a continuous basis.